

Submission A solution to Problem 2

The infimum is $\sqrt{3}$

Let $M_\beta = \Sigma / \langle G_\beta \rangle$. This is a compact flat Mobius band, with boundary a circle of length 2β . If β is realized, condition (2) says exactly that the given map is constant on the G_β -orbits and separates distinct orbits. Hence it descends to a continuous injection $F : M_\beta \rightarrow \mathbb{R}^3$, and this injection is an embedding because M_β is compact and $\mathbb{R}^3$ is Hausdorff. The locally finite invariant triangulation descends to a locally finite triangulation of the compact surface M_β , hence to a finite one.

In a clean triangle, call the boundary edge the ridge and the opposite vertex the apex. A bend is a segment in the triangle joining the apex to a point of the ridge.

Lemma 1 (bends lengthen). *Let L be the affine map on one clean triangle. If P is any point of the ridge and C is the apex, then*

$$|L(P) - L(C)| > |P - C|.$$

Moreover every non-degenerate boundary subarc is strictly shortened by F .

Proof. Let the ridge endpoints be A, B , and put $P_s = (1 - s)A + sB$. The function

$$q(s) = |L(P_s) - L(C)|^2 - |P_s - C|^2$$

satisfies

$$q''(s) = 2(|L(A) - L(B)|^2 - |A - B|^2) < 0,$$

because the ridge is shortened. Since the two side edges are lengthened, $q(0) > 0$ and $q(1) > 0$. Concavity gives $q(s) > 0$ for $0 \leq s \leq 1$. On each boundary edge the affine map has constant Lipschitz factor < 1 ; an arbitrary boundary subarc is split at the finitely many boundary vertices it meets, and the strict inequalities add. $\square$

Lemma 2 (a polygonal T -pair). *For $F(M_\beta)$ there are two bends whose image segments lie on perpendicular intersecting lines.*

Proof. The bends form a circle. Indeed, every bend meets the center circle $\{y = 1/2\} / \langle G_\beta \rangle$ once, and every point of the center circle lies in a unique bend; on a common non-boundary edge the two adjacent triangles give the same limiting bend.

Parametrize this bend circle by $u \in \mathbb{R}/\mathbb{Z}$, lift to $u \in \mathbb{R}$, and choose the image direction $e(u)$ from the lower side of the lifted strip to the upper side. Let $m(u)$ be the image midpoint. The previous lemma makes every image bend non-degenerate, and the finite polygonal structure gives continuous functions e, m . Going once around the Mobius band reverses the chosen orientation, so

$$e(u+1) = -e(u), \quad m(u+1) = m(u).$$

For $0 < r < 1$ define

$$\Phi(u, r) = \left(e(u) \cdot e(u+r), (m(u) - m(u+r)) \cdot (e(u) \times e(u+r)) \right).$$

It is periodic in u . As $r \rightarrow 0$ it tends to $(1, 0)$, and as $r \rightarrow 1$ it tends to $(-1, 0)$. Thus it extends continuously to the two-point compactification of $S^1 \times (0, 1)$, obtained by collapsing the two boundary circles of $S^1 \times [0, 1]$; this compactification is S^2 .

The involution $\iota(u, r) = (u+r, 1-r)$ interchanges the two bends. Under the coordinates

$$(\sin \pi r \cos 2\pi(u+r/2), \sin \pi r \sin 2\pi(u+r/2), \cos \pi r)$$

on the compactified cylinder, ι is the antipodal map. From $e(u+1) = -e(u)$ and $m(u+1) = m(u)$ one checks that $\Phi \circ \iota = -\Phi$. Borsuk–Ulam gives a zero of Φ .

At such a zero the two direction vectors are perpendicular. The second coordinate says that the vector joining the two midpoints lies in the plane spanned by those directions; hence the two supporting lines are coplanar. Since they are not parallel, they intersect. $\square$

Proposition 1. *Every realized β satisfies $\beta > \sqrt{3}$.*

Proof. Choose the two bends from Lemma 2. Their image segments have disjoint relative interiors, because F is an embedding. Let O be the intersection of their supporting lines. At least one image segment does not contain O in its relative interior; call that bend B , and call the other one T .

Lift T to a segment from $(0, 0)$ to $(t, 1)$. Replacing x by $-x$, if necessary, we may assume $t \geq 0$. In fact $t < \beta$: if $t > \beta$, this lifted segment meets its G_β -translate in its interior, and if $t = \beta$, its endpoints are G_β -equivalent. Either case contradicts the fact that the image of a bend is a non-degenerate affine segment and that F is embedded on the quotient.

Cutting M_β along T gives two boundary arcs H, D of lengths

$$\text{length}(H) = \beta - t, \quad \text{length}(D) = \beta + t. \tag{1}$$

Indeed the upper endpoint $(t, 1)$ has boundary coordinate $t + \beta$ modulo 2β . Let $T^* = F(T)$. Since $H^* = F(H)$ joins the endpoints of T^* , while boundary arcs are shortened and bends are lengthened,

$$\beta - t = \text{length}(H) > \text{length}(H^*) \geq |T^*| > \sqrt{1+t^2}. \tag{2}$$

The line of B^* is perpendicular to the line of T^* . Since O is not in the relative interior of B^* , one endpoint v^* of B^* has distance from the line of T^* at least $|B^*|$. The domain length

of B is at least 1, so Lemma 1 gives $|B^*| > 1$. Let A be the boundary arc, either H or D , which contains the corresponding boundary endpoint v .

The image arc A^* goes from one endpoint of T^* to the other through v^* . If a point has perpendicular distance h from a line segment of length L , the sum of its distances to the two endpoints is at least $\sqrt{L^2 + 4h^2}$, with equality only over the midpoint. Here $L = |T^*| > \sqrt{1+t^2}$ and $h > 1$, so

$$\text{length}(A) > \text{length}(A^*) \geq (\text{that sum}) > \sqrt{5+t^2}. \quad (3)$$

If $A = H$, then $D \geq H$ and (3) gives

$$2\beta = H + D > 2\sqrt{5+t^2} > 2\sqrt{3}.$$

Thus assume $A = D$. From (1)–(3),

$$2\beta = H + D > \sqrt{1+t^2} + \sqrt{5+t^2}, \quad (4)$$

and also

$$2\beta = 2D - 2t > 2\sqrt{5+t^2} - 2t. \quad (5)$$

Let

$$a(t) = \sqrt{1+t^2} + \sqrt{5+t^2}, \quad b(t) = 2\sqrt{5+t^2} - 2t.$$

On $[0, \infty)$, a is increasing and b is decreasing, and

$$a(1/\sqrt{3}) = b(1/\sqrt{3}) = 2\sqrt{3}.$$

Hence $\max(a(t), b(t)) \geq 2\sqrt{3}$ for all $t \geq 0$. Equations (4)–(5) imply $2\beta > 2\sqrt{3}$. $\square$

It remains to construct realizations for all larger β .

Lemma 3 (an embedded six-triangle model). *Let $E_0E_1E_2$ be an equilateral triangle of side a . For every sufficiently small $\varepsilon > 0$ there are points $C_i(\varepsilon)$ tending to the midpoint of E_iE_{i+1} (indices mod 3) such that the six affine triangles*

$$(E_i, C_i, E_{i+2}), \quad (C_i, E_{i+1}, E_{i+2}), \quad i = 0, 1, 2,$$

form an embedded polyhedral Mobius band with boundary $E_0C_0E_1C_1E_2C_2E_0$.

Proof. Embeddedness is preserved by nonsingular affine changes of coordinates, so it is enough to verify one model. Take

$$\begin{aligned} A_0 &= (0, 0, 0), & A_1 &= (1, 0, 0), & A_2 &= (0, 1, 0), \\ B_0 &= (1/2, 0, \varepsilon), & B_1 &= (1/2 - \varepsilon, 1/2, -2\varepsilon), & B_2 &= (\varepsilon, 1/2, 0), \end{aligned}$$

where $0 < \varepsilon < 1/4$, and take the six faces

$$\Delta_0 = A_0B_0A_2, \quad \Delta_1 = B_0A_1A_2, \quad \Delta_2 = A_1B_1A_0, \quad \Delta_3 = B_1A_2A_0,$$

$$\Delta_4 = A_2 B_2 A_1, \quad \Delta_5 = B_2 A_0 A_1.$$

The links of the vertices are intervals. For instance

$$\text{lk}(A_i) = B_i - A_{i+2} - B_{i+1} - A_{i+1} - B_{i+2}, \quad \text{lk}(B_i) = A_i - A_{i+2} - A_{i+1}$$

(indices mod 3, and reversal of a path is immaterial). Thus the abstract complex is a connected compact surface. Its single boundary cycle is $A_0 B_0 A_1 B_1 A_2 B_2 A_0$, and $\chi = 6 - 12 + 6 = 0$; hence, by the classification of compact surfaces, it is a Mobius band once its geometric realization is embedded.

It remains to check embeddedness. The face normals are

$$\begin{aligned} n_0 &= (-\varepsilon, 0, 1/2), & n_1 &= (\varepsilon, \varepsilon, 1/2), & n_2 &= (0, 2\varepsilon, 1/2), \\ n_3 &= (2\varepsilon, 0, 1/2 - \varepsilon), & n_4 &= (0, 0, 1/2 - \varepsilon), & n_5 &= (0, 0, 1/2). \end{aligned}$$

For the edge-sharing pairs $(0, 1), (0, 3), (1, 4), (2, 3), (2, 5)$ these normals are not proportional, so the two planes meet in the line of the common edge and the two triangles meet exactly in that edge. The remaining edge-sharing pair is $(4, 5)$. These two triangles are coplanar in $z = 0$, but their third vertices lie on opposite sides of their common edge $B_2 A_1$, since

$$\det(A_1 - B_2, A_2 - B_2) = \frac{1 - 2\varepsilon}{2} > 0, \quad \det(A_1 - B_2, A_0 - B_2) = -\frac{1}{2} < 0.$$

Therefore $\Delta_4 \cap \Delta_5 = B_2 A_1$.

The other nine pairs of faces share only one vertex. For each of them, the normals listed above are not proportional, so the two planes meet in a line through that vertex. Expressing one direction of this line in the two edge bases at the shared vertex gives the following sign table:

(i, j)	Δ_i	Δ_j
$(0, 2)$	$(-, +)$	$(-, +)$
$(0, 4)$	$(-, 0)$	$(-, +)$
$(0, 5)$	$(0, -)$	$(-, +)$
$(1, 2)$	$(-, +)$	$(+, -)$
$(1, 3)$	$(+, -)$	$(-, +)$
$(1, 5)$	$(0, +)$	$(+, -)$
$(2, 4)$	$(0, -)$	$(+, -)$
$(3, 4)$	$(0, -)$	$(-, +)$
$(3, 5)$	$(0, +)$	$(+, -)$

for $0 < \varepsilon < 1/4$. For example, for the pair $(0, 2)$ one line direction is $(-\varepsilon, \varepsilon/2, -2\varepsilon^2)$, with cone coordinates $(-2\varepsilon, \varepsilon/2)$ and $(\varepsilon(2\varepsilon - 3)/2, \varepsilon)$. The other entries are obtained by the same two-by-two linear calculation; zeros occur only where displayed. Call a sign pair non-negative (respectively non-positive) when both of its entries are so. In no row are the two sign pairs both non-negative or both non-positive. Therefore neither direction of the intersection line lies in both closed tangent cones, and the two closed triangles meet only at their common vertex. This proves the model is embedded. Applying the affine map which sends A_i to E_i gives the stated points $C_i(\varepsilon)$, and they tend to the side midpoints as $\varepsilon \rightarrow 0$. $\square$

Lemma 4 (realizations above $\sqrt{3}$). *Every $\beta > \sqrt{3}$ is realized.*

Proof. Put $\delta = \beta/3$. Choose a with

$$\max\{2/\sqrt{3}, \sqrt{1+\delta^2}\} < a < 2\delta;$$

such an a exists because $\delta > 1/\sqrt{3}$.

Triangulate M_β in the fundamental rectangle $[0, \beta] \times [0, 1]$, with $(0, y) \sim (\beta, 1 - y)$. Put

$$\begin{aligned} A_0 &= (0, 0), & B_0 &= (\delta, 0), & A_1 &= (2\delta, 0), & B_1 &= (\beta, 0), \\ B_1 &= (0, 1), & A_2 &= (\delta, 1), & B_2 &= (2\delta, 1), & A_0 &= (\beta, 1). \end{aligned}$$

Split the three rectangles of width δ by the diagonals A_0A_2 , A_1A_2 , and A_1A_0 , respectively. Equivalently, the six faces are

$$(A_i, B_i, A_{i+2}), \quad (B_i, A_{i+1}, A_{i+2}), \quad i = 0, 1, 2$$

(indices mod 3). This is a clean triangulation of M_β , and its lift to Σ is G_β -invariant.

Use Lemma 3 with side length a , and choose ε so small that the following strict inequalities, which hold in the limit $\varepsilon \rightarrow 0$, still hold:

$$|E_i C_i| < \delta, \quad |C_i E_{i+1}| < \delta, \quad |C_i E_{i+2}| > 1.$$

Also $|E_i E_j| = a > \sqrt{1+\delta^2}$. Map the six vertices A_i, B_i to E_i, C_i , respectively, and extend affinely over each of the six triangles. Lemma 3 gives an embedding $F : M_\beta \rightarrow \mathbb{R}^3$.

Each ridge in the domain has length δ , and its image has length $< \delta$. Each non-boundary edge is of one of two types: either it has domain length 1, and image length $|C_i E_{i+2}| > 1$, or it has domain length $\sqrt{1+\delta^2}$, and image length $a > \sqrt{1+\delta^2}$. Thus F is a squeeze map on every triangle. Lifting F to Σ gives a continuous map invariant under G_β ; since F is an embedding of the quotient, the lifted map satisfies condition (2). Hence β is realized. $\square$

Proposition 1 shows that realized numbers are all $> \sqrt{3}$, while Lemma 4 realizes every $\beta > \sqrt{3}$. Therefore the infimum of the set of realized numbers is $\sqrt{3}$.